

Risk Stratification Model Card and Feature Importance (Version 2025-10-01)

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1. Introduction

This document provides a comprehensive overview of the Risk Stratification Model and associated features utilized during the firmware impact period spanning from September 1, 2025, to November 10, 2025. The primary purpose of this model card is to offer transparency into the features

contributing to patient risk assessments, their relative importance, and the operational procedures for implementing and troubleshooting the risk scoring process in clinical settings.

It includes detailed descriptions of the model's computation methodology, feature importance rankings, relevant device firmware updates, vendor bulletins, error handling protocols, and decision policies guiding clinician outreach and resource allocation. This information supports compliance, quality assurance, and operational decision-making aligned with clinical workflows.

2. Collection Context

The documentation pertains to playbooks and firmware notes issued during the specified impact window. The data collection integrates multiple sources including device telemetry, clinician input, and automated risk scores. These sources collectively inform the risk stratification model, which is implemented within the clinical decision support system.

The context emphasizes ongoing device updates, feature capture accuracy, and the temporal relevance of model inputs in supporting timely interventions for high-risk patients. It also encompasses policies on data integrity, patient privacy, and compliance with relevant healthcare regulations.

3. Playbooks and Firmware Notes (2025-09-01 to 2025-11-10)

This section documents the operational playbooks and firmware updates issued during the specified period:

- Version upgrades, bug fixes, and feature enhancements for device firmware.
- Strategic workflows for deployment and testing of firmware updates.
- Real-world scenarios illustrating firmware impact on data collection and risk scoring.

4. Device Vendor Bulletins

4.1. DEVCO-GM v3.9

The firmware update DEVCO-GM v3.9 introduced the following changes:

- Enhanced Bluetooth transmission stability.
- Optimized sensor calibration algorithms.
- Added support for new glucose monitoring features.

4.2. DEVCO-BP v2.7

Firmware version DEVCO-BP v2.7 includes:

- Resolution of previous connectivity bugs.
- Improved battery life management.
- Expanded error logging capabilities for troubleshooting.

5. Error Codes and Troubleshooting

5.1. BT-GM-39xx Series

Error Code	Symptoms	Root Cause	Resolution Steps	Prevention Tips
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BT-GM-3901	Inconsistent glucose readings; device disconnects frequently during Bluetooth pairing	Sensor calibration drift or Bluetooth interference	1. Recalibrate sensor following protocol XYZ. 2. Ensure Bluetooth environment is free from interference sources. 3. Re-pair device following pairing instructions.	Perform periodic calibration and avoid placement near wireless interference sources.
BT-GM-3902	Failed data transmission logs or no data received	Firmware buffer overflow or connectivity loss	1. Restart device and firmware reset if needed. 2. Check Bluetooth signal strength and reconnect. 3. Update firmware to version 3.9 if not already installed.	Regular firmware updates and proper device placement.

5.2. BT-BP-27xx Series

Similar structure can be used for BP device error codes with specific symptoms, root causes, and resolutions listed accordingly.

6. Ingestion Troubleshooting Procedures

6.1. Common Ingestion Failures

Failures in data ingestion can stem from network issues, incompatible firmware versions, or data format mismatches. Troubleshooting involves the following steps:

- 1. Verify device firmware version and update if outdated.
- 2. Check network connectivity and firewall settings blocking data ports.
- 3. Validate data format compliance with ingestion API specifications.
- 4. Use diagnostic tools to assess connection logs.

6.2. Technical Checklist

Check Point	Expected Result	Action
Device Firmware Version	v3.9 for GM, v2.7 for BP or higher	Update firmware if below thresholds
Network Status	Stable connection; no packet loss	Troubleshoot network routing or hardware issues
Data Format	Match API schema version 5.4	Validate payload schemas against specification

7. Clinician Outreach Scripts

The following scripts are used to guide clinicians during patient engagement based on risk scores:

7.1. General Outreach Protocol

"Hello, this is a reminder that your recent health data indicates a potential risk. We recommend scheduling a check-up with your healthcare provider. Do you need assistance in booking an appointment or additional information?"

7.2. Automated Alert Trigger

When a patient's fasting glucose exceeds 140 mg/dL, an automated message prompts: *"Your recent test indicates elevated glucose levels. Please contact your healthcare provider for a follow-up."*

7.3. Follow-up Guidance

For patients with persistent high risk, clinicians are advised to perform comprehensive assessment and reinforce lifestyle modifications or medication adjustments as per policy.

8. Next Best Action Policy Thresholds

This policy defines threshold criteria for initiating clinical actions based on model scores and feature metrics:

- Immediate alert sent to nursing staff to review patient status.
- Initiates clinician review and potential medication adjustment.
- Active outreach to patient for device checkup or replacement.

Parameter	Threshold	Action Triggered	Description
Fasting Glucose	> 140 mg/dL	Nurse Check-In	
Risk Score (from model)	> 0.7 (scale 0-1)	Clinical Intervention	

Recent Device Errors	More than 2 errors in last 24 hours	Device Maintenance Call
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9. Risk Stratification Model Card

9.1. Model Overview

The risk stratification model predicts the likelihood of adverse health events based on multi-modal feature data collected during patient monitoring. It is designed to support timely intervention, optimize resource allocation, and improve patient outcomes.

The model employs gradient boosting algorithms trained on anonymized datasets, incorporating a wide array of clinical, device, and behavioral features.

9.2. Key Features Used

- Fasting glucose level
- Recent insulin dose adjustments
- Number of device errors in past 24 hours
- Sensor calibration consistency
- Patient activity levels (step count)
- Medication adherence indicators
- Previous hospitalizations

9.3. Model Deployment Context

The model operates in a real-time environment with data inputs refreshed every 15 minutes. During the firmware impact window, particular focus was on the influence of firmware version updates on data quality and feature reliability.

10. Feature Importance and Scoring Methodology

10.1. Feature Importance Rankings

Based on SHAP (SHapley Additive exPlanations) analysis, the most influential features in the current model version are:

Feature	Importance Score	Contribution Explanation
Fasting glucose	0.35	Strong predictor of hyperglycemia episodes.
Device error count	0.20	Indicates data reliability and recent instability.
Activity level (steps)	0.15	Correlated with metabolic health.
Medication adherence score	0.10	Reflects compliance impacting risk.
Sensor calibration variance	0.08	Impacts measurement accuracy.

10.2. Score Computation Process

The risk score is computed through a weighted aggregation of features as follows:

```
risk_score = sigmoid(Σ (weight_i * feature_value_i) + bias)
```

Where *weight_i* are feature-specific weights derived from model training, and *bias* is the intercept term. The sigmoid function ensures the score falls between 0 and 1.

Example calculation:

Given:

- Fasting Glucose = 150 mg/dL
- Device Error Count = 3
- Activity Level = 3000 steps
- Medication adherence score = 0.8
- Calibration Variance = 0.05

Resulting risk score $\approx$ 0.82

11. Clinical Note Generator QA Guidance

The Clinical Note Generator automates documentation based on risk and feature data. QA guidelines ensure accuracy, consistency, and compliance:

- Verify that generated notes accurately reflect recent data inputs and risk scores.
- Validate that standard templates incorporate correct terminology and clinical thresholds.
- Cross-check generated notes against manual documentation for sample cases.
- Ensure data privacy protocols are adhered to during QA reviews.

Sample QA Checklist:

1. The note includes patient's latest fasting glucose and risk score.
2. Interventions or advisories align with model policy thresholds.
3. No sensitive or identifiable patient information is inadvertently included.

Example of QA-verified note:

Patient's recent fasting glucose: 150 mg/dL indicates elevated